Food Expiry and Nutrition Tracking System

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About The Project

The Food Expiry and Nutrition Tracking System is a management solution developed to handle daily kitchen operations and inventory oversight. It enables the staff to systematically monitor food batches, actively track available quantities and identify items approaching their expiration dates to reduce waste. Furthermore, the platform facilitates detailed nutritional profiling for raw ingredients and features a recipe builder that automatically calculates the macronutrient content of menu items, seamlessly linking these culinary creations to an efficient customer order tracking module for a restaurant.

Work Done

Gauraang Shrivastava developed -

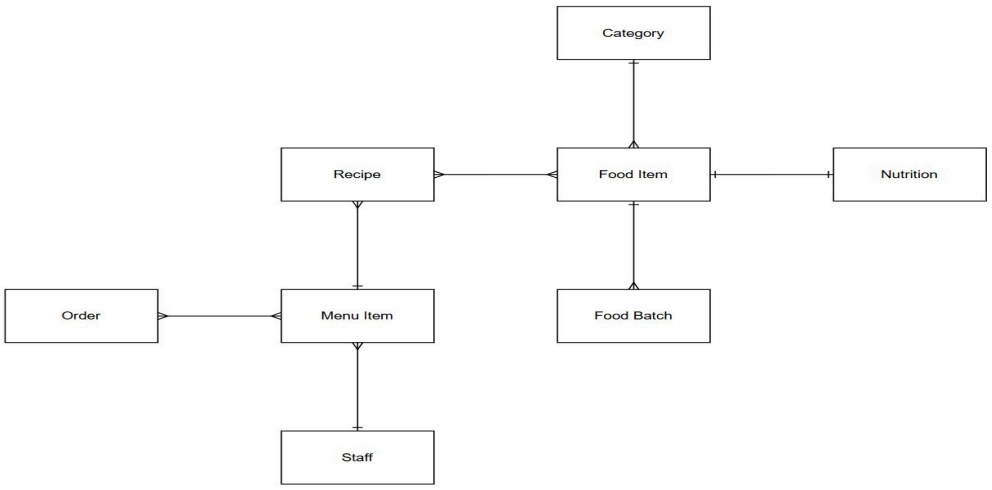
Front-end user interface
CSS styling,
ER diagram for the database schema.

Shreynash Singh developed-

Implemented JavaScript
Local Database Storage
Calculations required to know the nutritional value, Expiry dates and Orders

Project Summary

The methodology involved designing a relational database schema using an ER diagram.
The frontend was built with HTML and CSS, with JavaScript handling the core application logic.
Data persistence was achieved using LocalStorage to store records as JSON arrays.
Finally, functional modules were integrated for expiration tracking, nutritional calculations, and order management.



Result & Outcome

The project resulted in a fully functional web application that successfully utilizes LocalStorage for client-side data management. The final system effectively automates inventory expiration tracking, dynamically calculates recipe macronutrients, and streamlines customer order workflows.
The system operates through four main modules accessed after staff authentication:-

- Users manage stock in the Inventory module by adding food batches with expiration dates, which automatically update to show days remaining or an expired status.
- The Nutrition module allows users to define custom ingredients with specific macronutrient profiles.
- In the Recipes module, staff combine these ingredients to create menu items, with the system automatically calculating the total nutritional value based on portion sizes.
- Finally, the Orders module enables staff to log customer requests for menu items and track them from pending to completion.

Future Work

Move from client-side LocalStorage to a centralized relational database like MySQL with a backend API for real-time multi-user synchronization . We can also Integrate barcode scanning for automated inventory logging and implement automated notifications for items near expiration dates.

References

1. developer.mozilla.org
2. w3schools.com
3. app.diagrams.net

Github

gshriv006/FoodExpiry-NutritionTracking